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Philip et al.

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(54) **COMPOSITE MATERIAL PHASE CHANGE  
MEMORY CELL**

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CPC ..... *H10N 70/231* (2023.02); *H10B 63/00* (2023.02); *H10N 70/021* (2023.02); *H10N 70/8613* (2023.02)

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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,859,893 B2 \* 12/2010 Liu ..... H10N 70/823 977/754  
7,973,384 B2 \* 7/2011 Happ ..... H10N 70/8828 257/536

(Continued)

FOREIGN PATENT DOCUMENTS

WO 2010097302 A1 9/2010

OTHER PUBLICATIONS

Brew et al., "Phase Change Memory Cell Sidewall Projection Liner," U.S. Appl. No. 17/489,602, filed Sep. 29, 2021.

(Continued)

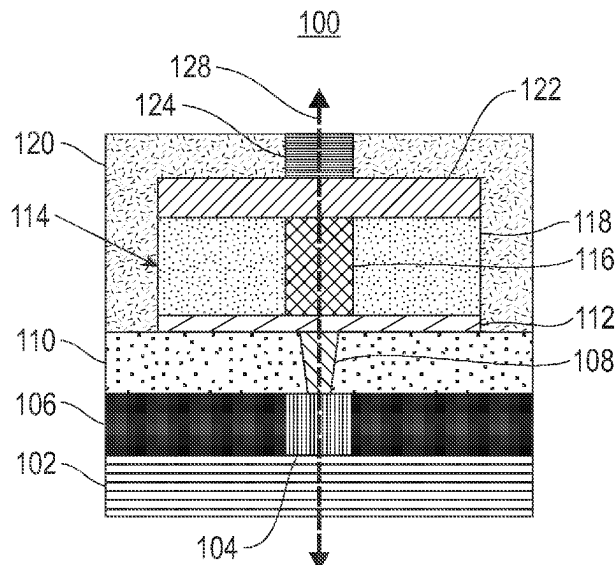
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(57) **ABSTRACT**

A phase change memory (PCM) cell includes a first electrode comprised of a first electrically conductive material, a second electrode comprised of a second electrically conductive material, and a phase change section positioned between the first electrode and the second electrode. The phase change section includes a first phase change material having a first resistance drift coefficient, and a second phase change material having a second resistance drift coefficient that is greater than the first resistance drift coefficient. An axis of the PCM cell extends between the first electrode and the second electrode, and the second phase change material is offset from the first phase change material in a direction that is perpendicular to the axis.

19 Claims, 8 Drawing Sheets



(56)

**References Cited**

## U.S. PATENT DOCUMENTS

|              |      |         |             |       |             |
|--------------|------|---------|-------------|-------|-------------|
| 8,124,950    | B2 * | 2/2012  | Happ        | ..... | H10N 70/023 |
|              |      |         |             |       | 257/3       |
| 8,164,949    | B2   | 4/2012  | Savransky   |       |             |
| 8,237,145    | B2 * | 8/2012  | Kamata      | ..... | H10N 70/826 |
|              |      |         |             |       | 977/734     |
| 9,257,639    | B2   | 2/2016  | Kim         |       |             |
| 9,558,823    | B1   | 1/2017  | Khwa        |       |             |
| 9,882,126    | B2 * | 1/2018  | BrightSky   | ..... | H10N 70/884 |
| 10,991,879   | B2   | 4/2021  | Ruiz        |       |             |
| 11,456,417   | B2 * | 9/2022  | Brew        | ..... | H10N 70/826 |
| 2020/0381618 | A1   | 12/2020 | Cappelletti |       |             |
| 2021/0257547 | A1 * | 8/2021  | Shen        | ..... | H10N 70/826 |

## OTHER PUBLICATIONS

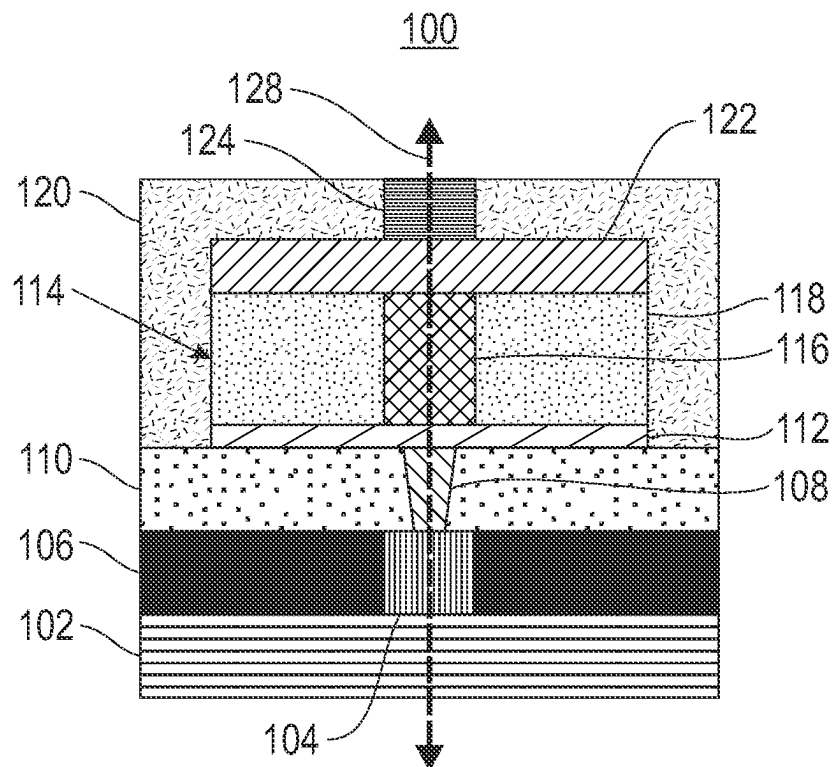
Ding et al., "Phase-change heterostructure enables ultralow noise and drift for memory operation," Science, vol. 366, Oct. 11, 2019, pp. 210-215, DOI: 10.1126/science.aay0291.

Kersting et al., "State dependence and temporal evolution of resistance in projected phase change memory," Scientific Reports, vol. 10, Article No. 8248, May 19, 2020, 11 pages. <[https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7237438/pdf/41598\\_2020\\_Article\\_64826.pdf](https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7237438/pdf/41598_2020_Article_64826.pdf)>.

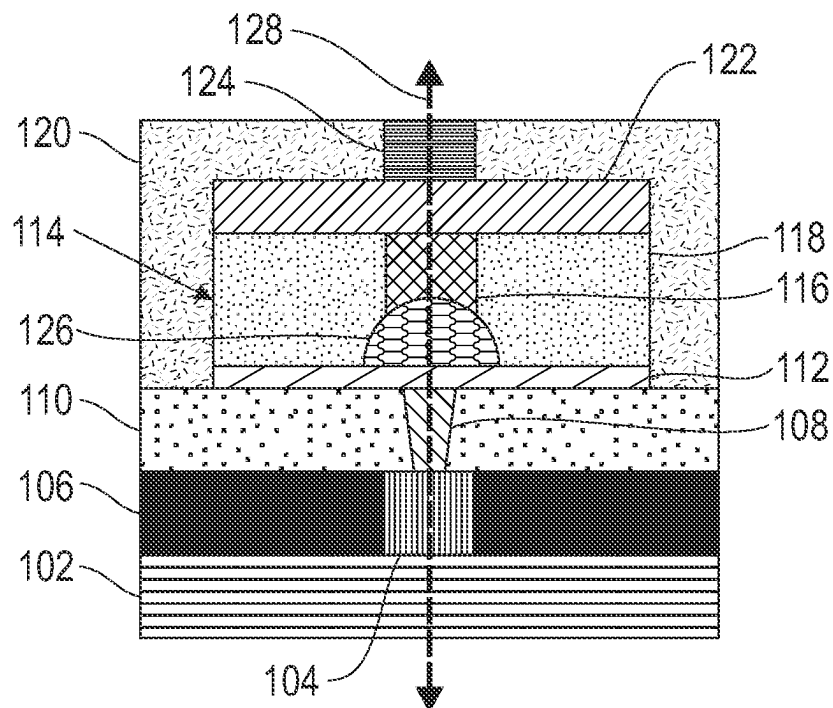
Shen et al., "Phase Change Memory Using Multiple Stacks of PCM Materials," U.S. Appl. No. 16/789,502, filed Feb. 13, 2020.

Simpson et al., "Interfacial phase-change memory," Nature Nanotechnology, vol. 6, Aug. 2011, pp. 501-505, DOI: 10.1038/nnano.2011.96.

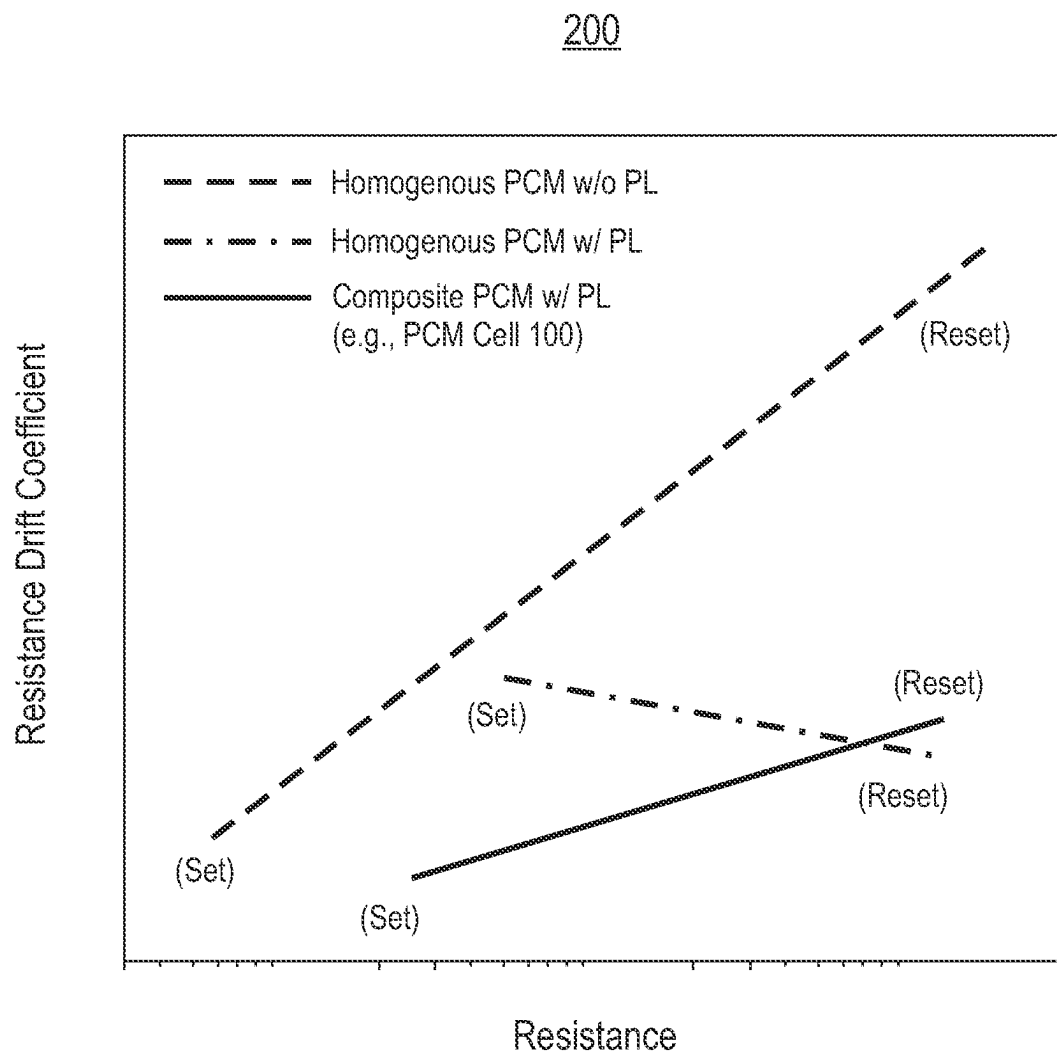
\* cited by examiner

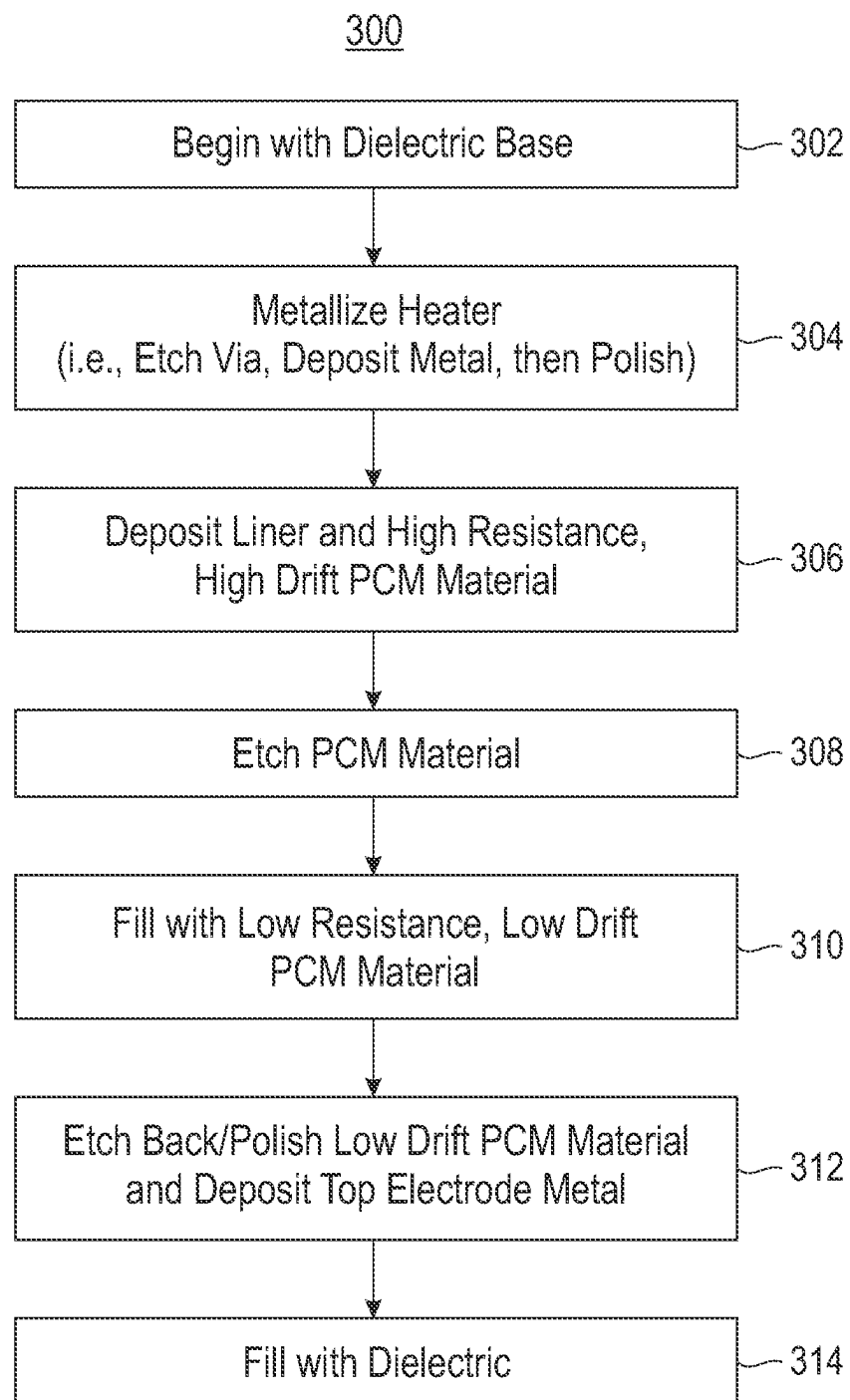


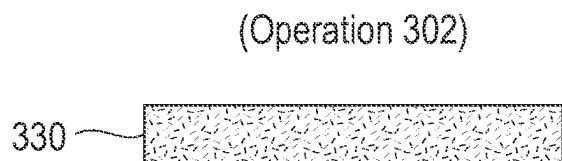
**FIG. 1A**



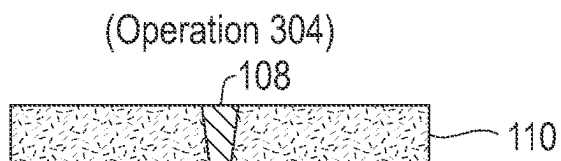
**FIG. 1B**

**FIG. 2**

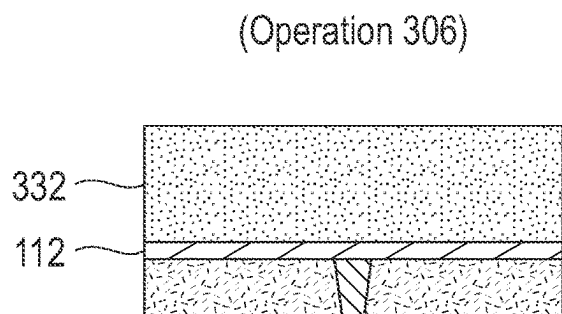




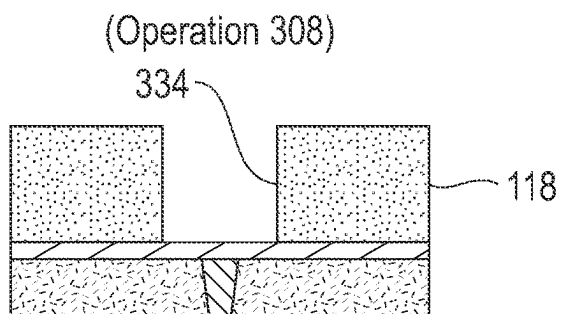
**FIG. 4A**



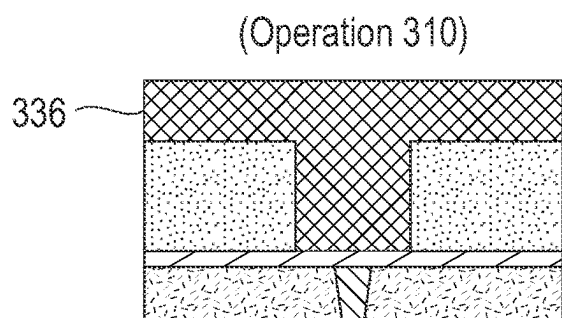
**FIG. 4B**



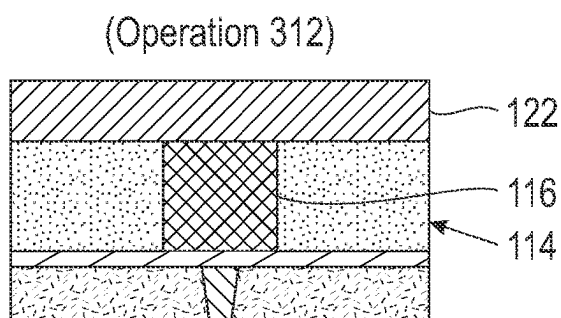
**FIG. 4C**



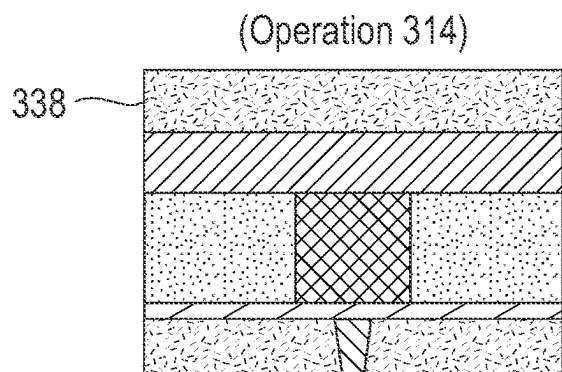
**FIG. 4D**



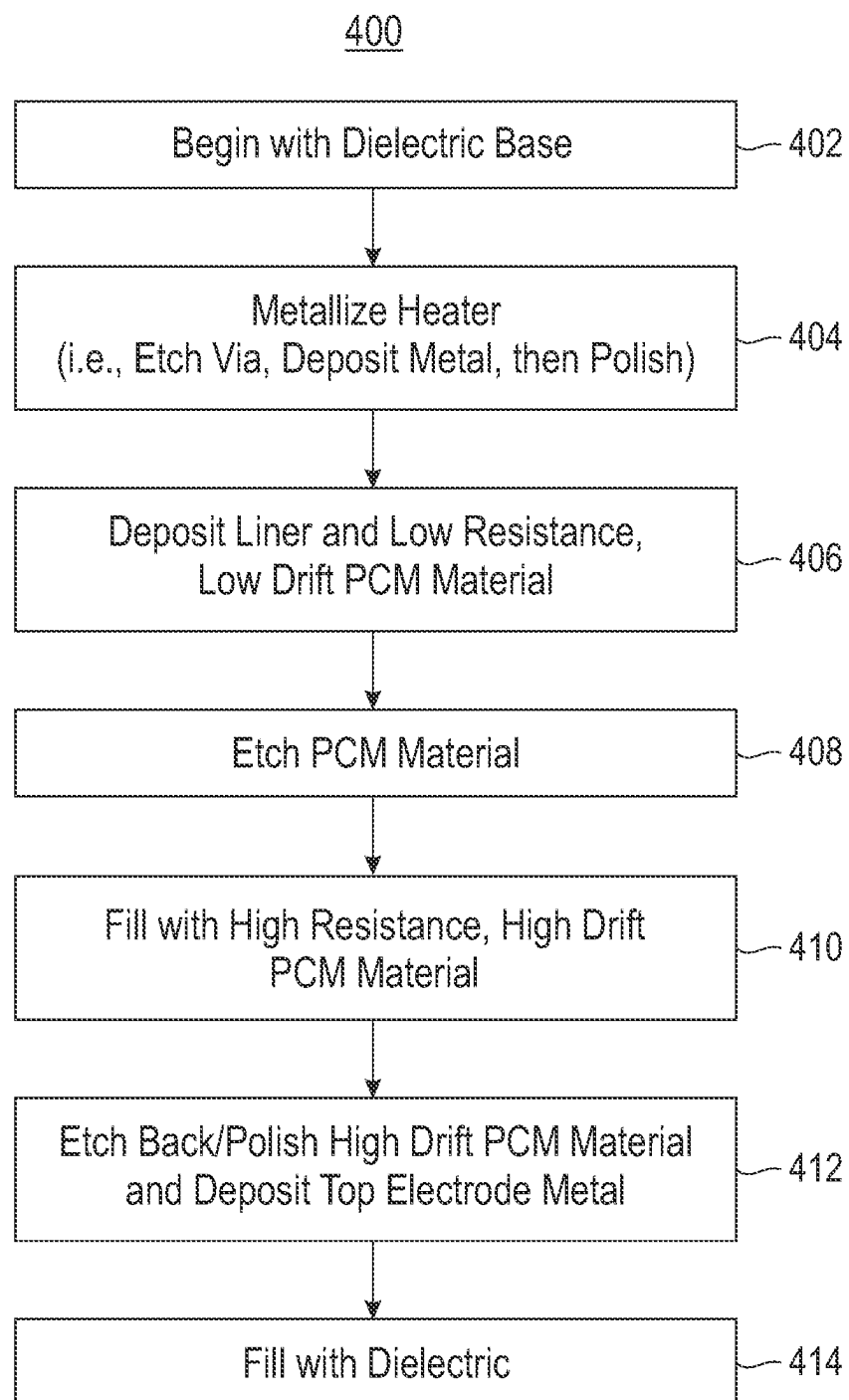
**FIG. 4E**



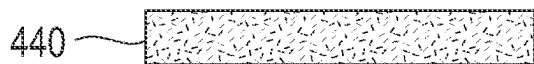
**FIG. 4F**



**FIG. 4G**

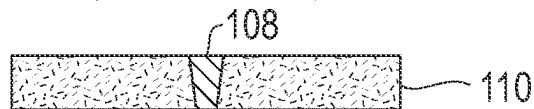


(Operation 402)



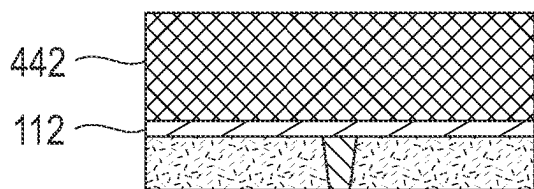
**FIG. 6A**

(Operation 404)



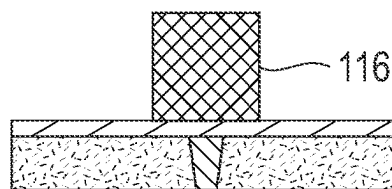
**FIG. 6B**

(Operation 406)



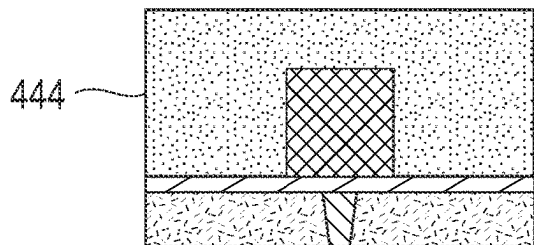
**FIG. 6C**

(Operation 408)



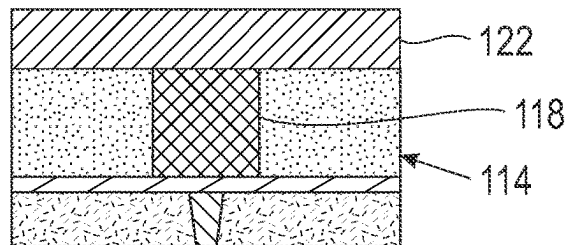
**FIG. 6D**

(Operation 410)



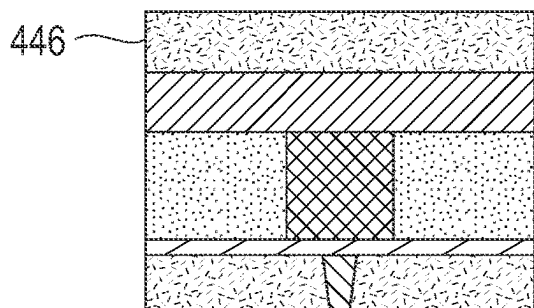
**FIG. 6E**

(Operation 412)



**FIG. 6F**

(Operation 414)



**FIG. 6G**



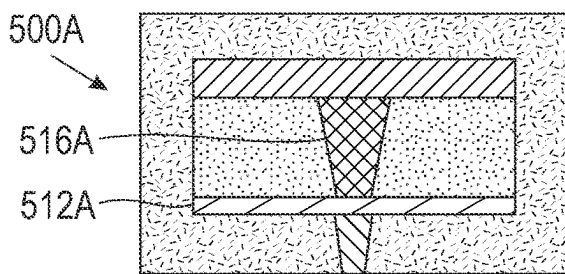


FIG. 7A

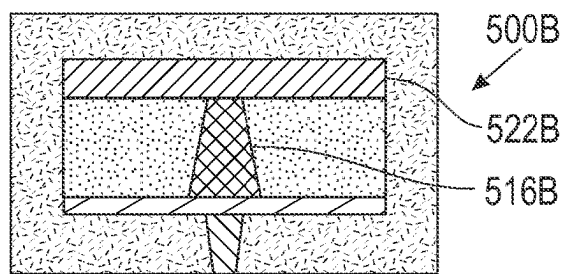


FIG. 7B

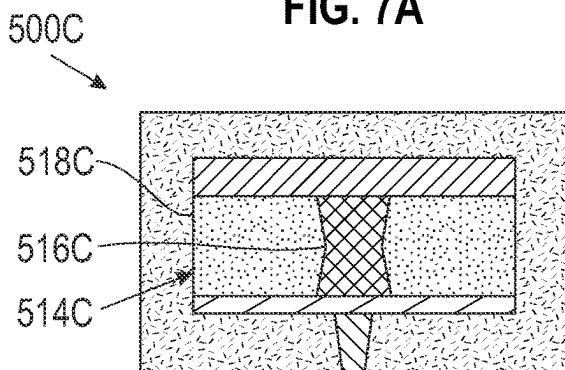


FIG. 7C

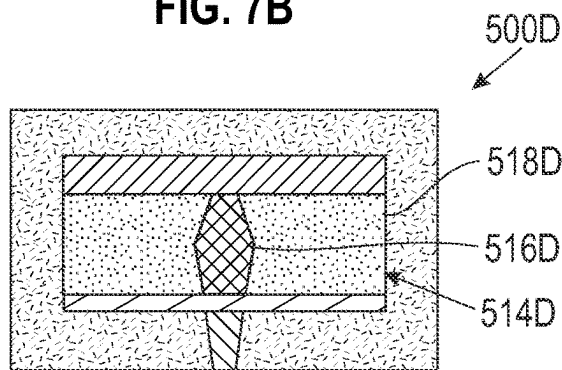


FIG. 7D

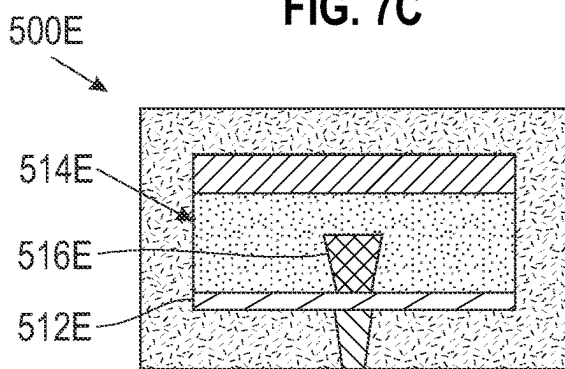


FIG. 7E

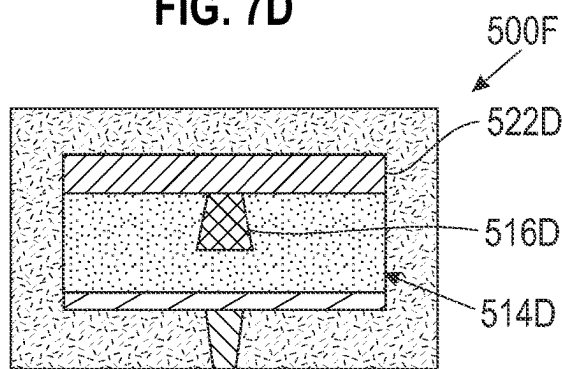


FIG. 7F

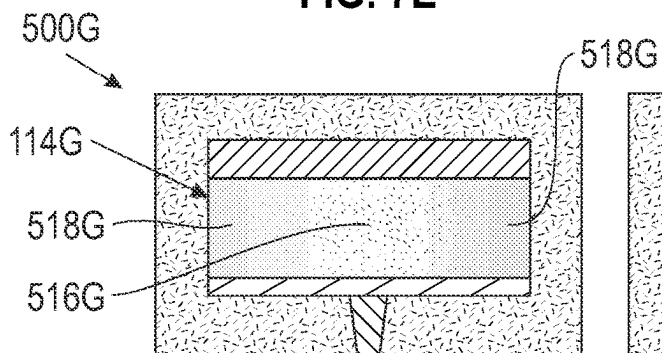


FIG. 7G

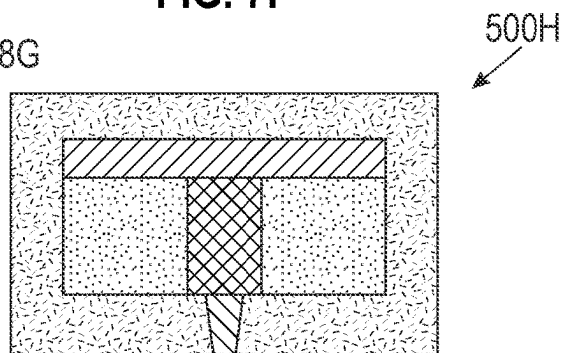


FIG. 7H

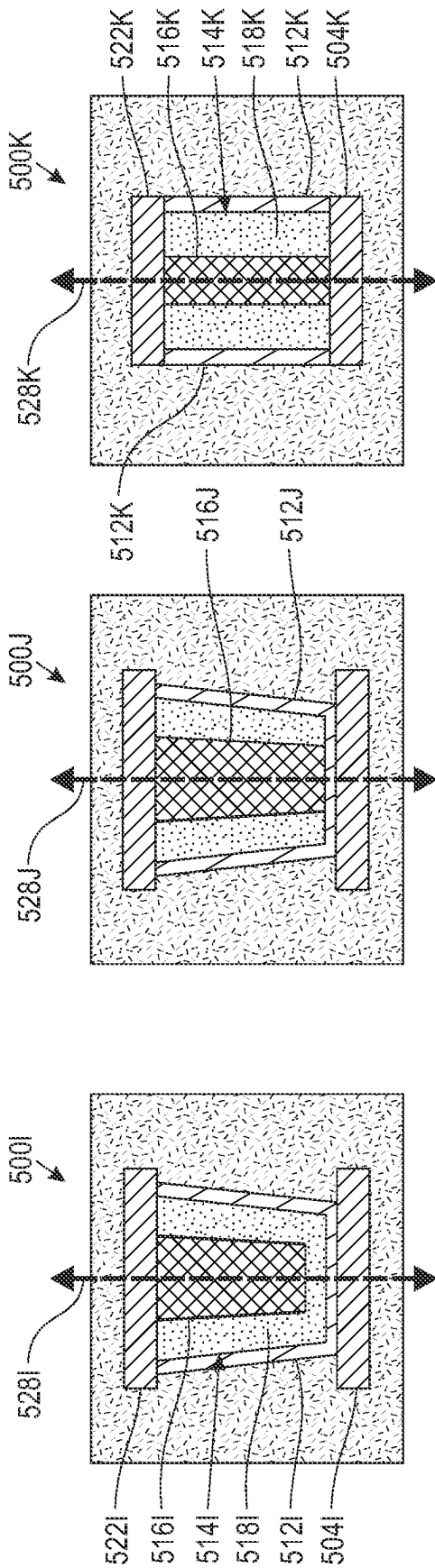


FIG. 8A

FIG. 8B

FIG. 9

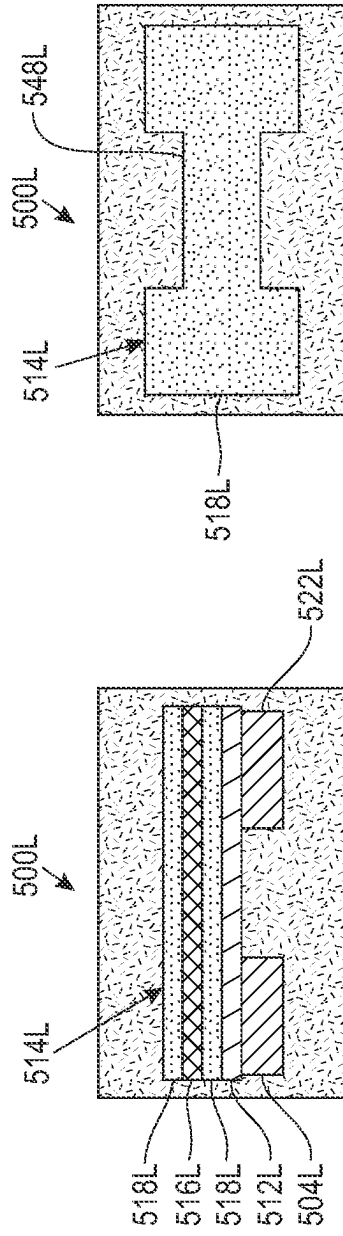


FIG. 10A

FIG. 10B

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## COMPOSITE MATERIAL PHASE CHANGE MEMORY CELL

### BACKGROUND

The present invention relates to computer memory, and more specifically, to phase change material memory devices with composite phase change materials.

Phase change memory (PCM) can be utilized for both training and inference in analog computing for artificial intelligence. The PCM structures can include phase change memristive devices with tunable conductivities and overall high device resistance with high retention to minimize energy consumption. The tuning can be accomplished by forming different structural states with varying proportions of crystalline and amorphous phases of PCM material. However, PCM materials can suffer from resistance drift over time, which can negatively affect the fidelity of the tuning.

### SUMMARY

According to an embodiment of the present disclosure, a PCM cell includes a first electrode comprised of a first electrically conductive material, a second electrode comprised of a second electrically conductive material, and a phase change section positioned between the first electrode and the second electrode. The phase change section includes a first phase change material having a first resistance drift coefficient, and a second phase change material having a second resistance drift coefficient that is greater than the first resistance drift coefficient. An axis of the PCM cell extends between the first electrode and the second electrode, and the second phase change material is offset from the first phase change material in a direction that is perpendicular to the axis.

According to an embodiment of the present disclosure, a method of manufacturing a PCM cell includes forming a first electrode, forming a phase change section electrically connected to the first electrode, and forming a second electrode on the phase change section. The phase change section includes a first phase change material having a first resistance drift coefficient, and a second phase change material having a second resistance drift coefficient that is greater than the first resistance drift coefficient.

According to an embodiment of the present disclosure, a PCM cell includes a first electrode comprised of a first electrically conductive material, a second electrode comprised of a second electrically conductive material, and a phase change section positioned between the first electrode and the second electrode. The phase change section includes a first phase change material having a first resistivity, and a second phase change material having a second resistivity that is greater than the first resistivity. An axis of the PCM cell extends between the first electrode and the second electrode, and the second phase change material is offset from the first phase change material in a direction that is perpendicular to the axis.

### BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a cross-section view of a mushroom PCM cell with a composite phase change section in a polycrystalline configuration, in accordance with embodiments of the present disclosure.

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FIG. 1B is a cross-section view of the mushroom PCM cell with the composite phase change section including an amorphous zone, in accordance with embodiments of the present disclosure.

FIG. 2 is a graph of resistance drift coefficient versus logarithmic resistance between the set and reset phases for various PCM cell configurations, in accordance with embodiments of the present disclosure.

FIG. 3 is a flowchart of a method of manufacturing the PCM cell of FIG. 1A, in accordance with embodiments of the present disclosure.

FIGS. 4A-4G are a series of cross-section views of the method of FIG. 3 of manufacturing the PCM cell, in accordance with embodiments of the present disclosure.

FIG. 5 is a flowchart of a method of manufacturing the PCM cell of FIG. 1A, in accordance with embodiments of the present disclosure.

FIGS. 6A-6G are a series of cross-section views of the method of FIG. 5 of manufacturing the PCM cell, in accordance with embodiments of the present disclosure.

FIGS. 7A-7H are cross-section views of alternate embodiment mushroom PCM cells, in accordance with embodiments of the present disclosure.

FIGS. 8A and 8B are cross-section views of alternate embodiment confined PCM cells, in accordance with embodiments of the present disclosure.

FIG. 9 is a cross-section view of alternate embodiment pillar PCM cell, in accordance with embodiments of the present disclosure.

FIG. 10A is a cross-section view of an alternate embodiment bridge PCM cell, in accordance with embodiments of the present disclosure.

FIG. 10B is a top view of bridge PCM cell of FIG. 10A, in accordance with embodiments of the present disclosure.

### DETAILED DESCRIPTION

Various embodiments of the present disclosure are described herein with reference to the related drawings. Alternative embodiments can be devised without departing from the scope of the present disclosure. It is noted that various connections and positional relationships (e.g., over, below, adjacent, etc.) are set forth between elements in the following description and in the drawings. These connections and/or positional relationships, unless specified otherwise, can be direct or indirect, and the present disclosure is not intended to be limiting in this respect. Accordingly, a coupling of entities can refer to either a direct or an indirect coupling, and a positional relationship between entities can be a direct or indirect positional relationship. As an example of an indirect positional relationship, references in the present description to forming layer "A" over layer "B" include situations in which one or more intermediate layers (e.g., layers "C" and "D") are between layer "A" and layer "B" as long as the relevant characteristics and functionalities of layer "A" and layer "B" are not substantially changed by the intermediate layer(s).

The following definitions and abbreviations are to be used for the interpretation of the claims and the specification. As used herein, the terms "comprises," "comprising," "includes," "including," "has," "having," "contains" or "containing," or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a composition, a mixture, process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but can include other elements not expressly listed or inherent to such composition, mixture, process,

method, article, or apparatus. In addition, any numerical ranges included herein are inclusive of their boundaries unless explicitly stated otherwise.

For purposes of the description hereinafter, the terms “upper,” “lower,” “right,” “left,” “vertical,” “horizontal,” “top,” “bottom,” and derivatives thereof shall relate to the described structures and methods, as oriented in the drawing figures. The terms “overlying,” “atop,” “on top,” “positioned on” or “positioned atop” mean that a first element, such as a first structure, is present on a second element, such as a second structure, wherein intervening elements such as an interface structure can be present between the first element and the second element. The term “direct contact” means that a first element, such as a first structure, and a second element, such as a second structure, are connected without any intermediary conducting, insulating or semiconductor layers at the interface of the two elements. It should be noted, the term “selective to,” such as, for example, “a first element selective to a second element,” means that a first element can be etched, and the second element can act as an etch stop.

For the sake of brevity, conventional techniques related to semiconductor device and integrated circuit (IC) fabrication may or may not be described in detail herein. Moreover, the various tasks and process steps described herein can be incorporated into a more comprehensive procedure or process having additional steps or functionality not described in detail herein. In particular, various steps in the manufacture of semiconductor devices and semiconductor-based ICs are well known and so, in the interest of brevity, many conventional steps will only be mentioned briefly herein or will be omitted entirely without providing the well-known process details.

In general, the various processes used to form a microchip that will be packaged into an IC fall into four general categories, namely, film deposition, removal/etching, semiconductor doping and patterning/lithography.

Deposition can be any process that grows, coats, or otherwise transfers a material onto the wafer. Available technologies include physical vapor deposition (PVD), chemical vapor deposition (CVD), electrochemical deposition (ECD), molecular beam epitaxy (MBE) and more recently, atomic layer deposition (ALD) among others. Another deposition technology is plasma enhanced chemical vapor deposition (PECVD), which is a process which uses the energy within the plasma to induce reactions at the wafer surface that would otherwise require higher temperatures associated with conventional CVD. Energetic ion bombardment during PECVD deposition can also improve the film's electrical and mechanical properties.

Removal/etching can be any process that removes material from the wafer. Examples include etch processes (either wet or dry), chemical mechanical planarization (CMP), and the like. One example of a removal process is ion beam etching (IBE). In general, IBE (or milling) refers to a dry plasma etch method which utilizes a remote broad beam ion/plasma source to remove substrate material by physical inert gas and/or chemical reactive gas means. Like other dry plasma etch techniques, IBE has benefits such as etch rate, anisotropy, selectivity, uniformity, aspect ratio, and minimization of substrate damage. Another example of a dry removal process is reactive ion etching (RIE). In general, RIE uses chemically reactive plasma to remove material deposited on wafers. With RIE the plasma is generated under low pressure (vacuum) by an electromagnetic field. High-energy ions from the RIE plasma attack the wafer surface and react with it to remove material.

Semiconductor doping can be the modification of electrical properties by doping, for example, transistor sources and drains, generally by diffusion and/or by ion implantation. These doping processes are followed by furnace annealing or by rapid thermal annealing (“RTA”). Annealing serves to activate the implanted dopants. Films of both conductors (e.g., poly-silicon, aluminum, copper, etc.) and insulators (e.g., various forms of silicon dioxide, silicon nitride, etc.) are used to connect and isolate transistors and their components. Selective doping of various regions of the semiconductor substrate allows the conductivity of the substrate to be changed with the application of voltage. By creating structures of these various components, millions of transistors can be built and wired together to form the complex circuitry of a modern microelectronic device.

Semiconductor lithography can be the formation of three-dimensional relief images or patterns on the semiconductor substrate for subsequent transfer of the pattern to the substrate. In semiconductor lithography, the patterns are formed by a light sensitive polymer called a photoresist. To build the complex structures that make up a transistor and the many wires that connect the millions of transistors of a circuit, lithography and etch pattern transfer steps are repeated multiple times. Each pattern being printed on the wafer is aligned to the previously formed patterns and gradually the conductors, insulators and selectively doped regions are built up to form the final device.

FIGS. 1A and 1B are cross-section views of PCM cell **100** for use in, for example, an integrated circuit (not shown). In the illustrated embodiment, PCM cell **100** comprises bottom wire **102**, bottom electrode **104**, insulator **106**, heater **108**, insulator **110**, projection liner (PL) **112**, PCM section **114** (which includes a first, undoped material **116** and a second, doped material **118**), insulator **120**, top electrode **122**, and top wire **124**.

In the illustrated embodiment, the bottom of bottom electrode **104** is in direct contact with and electrically connected to the top of bottom wire **102**, which can receive electrical signals from other components (not shown) of the integrated circuit. The bottom of heater **108** is in direct contact with and electrically connected to the top of bottom electrode **104**. The bottom of PL **112** is in direct contact with and electrically and thermally connected to the top of heater **108**. The bottom of PCM section **114** is in direct contact with and electrically and thermally connected to the top of PL **112**. The bottom of top electrode **122** is in direct contact with and electrically connected to the top of PCM section **114**. The bottom of top wire **124** is in direct contact with and electrically connected to the top of top electrode **122**, and top wire **124** can deliver electrical signals from PCM cell **100** to other components (not shown) of the integrated circuit.

In the illustrated embodiment, insulators **106**, **110**, **120** structurally support and electrically isolate the other components of PCM cell **100**, selectively, and fill in the space therebetween, as appropriate. Thus, the outer side of bottom electrode **104** is in direct contact with and laterally surrounded by insulator **106**, and the outer side of heater **108** is in direct contact with and laterally surrounded by insulator **110**. Furthermore, the bottom side of PL **112** is in direct contact with and axially adjacent to insulator **110**, and top wire **124** are in direct contact with and laterally surrounded by insulator **120**, and top electrode **122** is in direct contact with and laterally and axially adjacent to insulator **120**.

In the illustrated embodiment, a cross-section of PCM cell **100** (into the page in FIG. 1) can be circular, although in other embodiments, it can be rectangular, square, oval, or

any other suitable shape. In addition, the widths of PCM section 114, PL 112, and top electrode 122 are the same, whereas the width of heater 108 is substantially reduced, comparatively (e.g., three to seven times smaller, or about five times smaller). Thereby, PCM cell 100 can be said to have a mushroom configuration wherein an electrical signal (i.e., electrical current) can flow from bottom electrode 104 to top electrode 122 through heater 108, PL 112, and PCM section 114. In some embodiments, the width of heater 108 is between 35 and 40 nanometers (nm), and the widths of PL 112, PCM section 114, and top electrode 122 are between 100 nm and 400 nm. PCM section 114 has a height of about 80 nm, and the width of undoped material 116 within PCM section 114 is between 30 nm and 60 nm. Thus, in some embodiments, the width of doped material 118 can be one-and-a-half to fourteen times larger than undoped material 116, and in some embodiments, the width of doped material 118 can be three to seven times larger than undoped material 116. Furthermore, in some embodiments, the width of undoped material 116 can be three-quarters to twice the size of heater 108.

In the illustrated embodiment, heater 108 and undoped material 116 are positioned centrally along axis 128 of PCM cell 100 that extends between bottom electrode 104 and top electrode 122 (which is parallel to the direction of current flow in PCM cell 100 during operation). As depicted, axis 128 may extend longitudinally through PCM cell 100 such that axis 128 extends through each of top wire 124, top electrode 122, undoped material 116, heater 108, bottom electrode 104, and bottom wire 102. In some embodiments, PCM cell 100 may be mirrored in at least some directions across axis 128. Doped material 118 surrounds undoped material 116, and doped material 118 is offset from axis 128 in a lateral direction (i.e., perpendicular to axis 128). Thereby, undoped material 116 is positioned closer to heater 108 than doped material 118 is.

In the illustrated embodiment, bottom electrode 104 and top electrode 122 are comprised of a very electrically conductive material, such as metal or metallic compound, for example, titanium nitride (TiN) or tungsten (W). Heater 108 is an electrode that is comprised of TiN or a higher resistance metal, such as, for example, titanium tungsten (TiW), tantalum nitride (Ta<sub>2</sub>N<sub>5</sub>), or titanium aluminide (TiAl), and has a relatively narrow cross-sectional area, which focuses electrical current that is run through PCM cell 100. This allows heater 108 to generate heat through resistive heating during a pulse of electricity, which can be used to selectively change the temperature of PCM section 114, for example, above the crystallization temperature and the melting temperature of undoped material 116 and doped material 118. In addition, heater 108 can be comprised of multiple different electrically conductive materials that can be arranged in multiple layers.

In the illustrated embodiment, insulators 106, 110, 120 are comprised of a dielectric (electrical insulating) material, such as, for example, silicon nitride (SiN), silicon oxide (SiO<sub>2</sub>), silicon nitride carbide (SiNC), or tetraethyl orthosilicate (TEOS). In some embodiments, all of the insulators 106, 110, 120 are the same material, and in other embodiments, different materials are used for some or all of insulators 106, 110, 120. In addition, PL 112 is comprised of a moderately electrically resistive material, such as a metal and/or semiconductor (e.g., TaN; tungsten nitride (WN); amorphous carbons (a-C); doped a-C; transparent conductive oxides such as tin-doped indium oxide (ITO), aluminum zirconium oxide (AZO), and high-resistance metal chalcogenides (ex. titanium selenide (TiSe)), and other poorly

conducting metal nitrides). The material comprising PL 112 have a higher electrical resistivity than polycrystalline phases of undoped material 116 and doped material 118 but lower electrical resistivity than amorphous phases undoped material 116 and doped material 118.

In the illustrated embodiment, undoped material 116 is composed essentially of a phase change material such as a germanium-antimony-tellurium (GST), gallium-antimony-tellurium (GaST), or silver-iridium-antimony-telluride (AIST) material, although other materials can be used as appropriate. Examples of other PCM materials can include, but are not limited to, germanium-tellurium compound material (GeTe), silicon-antimony-tellurium (Si—Sb—Te) alloys, gallium-antimony-tellurium (Ga—Sb—Te) alloys, germanium-bismuth-tellurium (Ge—Bi—Te) alloys, indium-tellurium (In—Se) alloys, arsenic-antimony-tellurium (As—Sb—Te) alloys, silver-indium-antimony-tellurium (Ag—In—Sb—Te) alloys, Ge—In—Sb—Te alloys, Ge—Sb alloys, Sb—Te alloys, Si—Sb alloys, Ge—Te alloys and combinations thereof. The terms “composed essentially” and “consist essentially,” as used herein with respect to materials of different layers, indicates that other materials, if present, do not materially alter the basic characteristics of the recited materials. For example, an undoped material 116 consisting essentially of GST material does not include other materials that materially alter the basic characteristics of the GST material.

On the other hand, doped material 118 can be a mixture of a phase change material (e.g., similar to or the same as undoped material 116) and a phase separated dopant material such as, for example, one or more dielectric materials and/or poorly-electrically conductive materials (e.g., oxygen (O), nitrogen (N), carbon (C), SiO<sub>2</sub>, SiO, SiON, SiOC, tantalum nitride (Ta<sub>3</sub>N<sub>5</sub>), aluminum nitride (AlN), and titanium nitride (TiN)). The grains of the phase separated dopant material can restrict the grain size of the phase change material and provide “nano opens” (i.e., local regions of relatively high electrical resistance) to increase the resistance of doped material 118, and, in some embodiments, the amount of the phase separated dopant material doped material 118 is at least 10% by volume. Doped material 118 can also be a substitutional or interstitial doped phase change material such as, for example, titanium-GST (TiGST), gallium-GST (GaGST), silicon-GST (SiGST), or bismuth-GST (BiGST), since these atoms can substitute/sit on interstices due to their solubility in GST.

In the illustrated embodiment, PCM cell 100 can be operated as a memory cell by passing an electrical current pulse from bottom electrode 104 to top electrode 122 to program PCM cell 100. This can be done at a variety of voltages and/or for a variety of durations to read or write a value on PCM cell 100. For example, to write, a high voltage can be used (e.g., 1 volt (V) to 4 V) for a short duration, which can cause heater 108 to locally heat PCM section 114 beyond the melting points of undoped material 116 and doped material 118. Once the flow of current ceases, PCM section 114 can cool down rapidly, which forms amorphous zone 126 in a process called “resetting”. Zone 126 is a dome-shaped region of PCM section 114 having an amorphous configuration (in both undoped material 116 and doped material 118), although the remainder of PCM section 114 is still in a polycrystalline configuration. In general, this amorphous configuration has no definite structure. However, there can be local, disjoint crystalline nuclei (i.e., small, crystallized regions of phase change section 114) present in zone 126. The creation of zone 126 can cause the electrical resistance across PCM cell 100 to increase as compared to

a solely polycrystalline configuration (a la PCM cell **100** in FIG. **1A**). These resistance values of PCM cell **100** can be read without changing the phase of PCM section **114** (including that of zone **126**) or the resistance value of PCM cell **100**, for example, by sending a current pulse at a low voltage (e.g., 0.2 V) from bottom electrode **104** to top electrode **122**.

In addition, PCM section **114** can be rewritten and returned back to a solely polycrystalline configuration by “setting” PCM cell **100**. One way to rewrite PCM section **114** uses a high voltage electrical pulse (e.g., 1 V to 4 V) for a short period of time (e.g., 10 nanoseconds (ns)), which can cause PCM section **114** to heat up beyond the crystallization points of undoped material **116** and doped material **118** but not to their melting points. Since the crystallization temperature is lower than the melting temperature, once the flow of current ceases, PCM section **114** can anneal and form crystals. Another way to rewrite PCM section **114** uses an electrical pulse with a relatively long trailing edge (e.g., 1 microsecond) (as opposed to a square pulse with a relatively short trailing edge on the order of nanoseconds) that is strong enough to heat PCM section **114** beyond the melting points of undoped material **116** and doped material **118**, after which, PCM section **114** is cooled down slowly, allowing crystals to form. Either of these processes cause the electrical resistance across PCM cell **100** to decrease as compared to having an amorphous zone **126**. This new resistance value can then be read using current at a low voltage (e.g., 0.2 V) without changing the phase of PCM section **114** or the resistance value of PCM cell **100**.

In some embodiments, the melting temperatures of undoped material **116** and doped material **118** can be around 600° C. In some embodiments, the crystallization temperatures of undoped material **116** and doped material **118** can be around 180° C. In addition, the process of setting and resetting PCM cell **100** can occur repeatedly, and in some embodiments, different zones **126** with different resistances can be created in PCM materials **114** (e.g., due to having different sizes of zone **126** and/or amounts of crystallization nuclei in zone **126**). This allows for PCM cell **100** to have various distinct resistances that can be created by varying the resetting parameters. Thereby, if PCM cell **100** is considered to represent information digits, these digits can be non-binary (as opposed to traditional bits). However, in some embodiments, PCM cell **100** can be used as a bit by either having or not having a uniform zone **126** in PCM section **114**. In such embodiments, PCM cells **100** can have a high resistance (a.k.a., low voltage output or “0”) or low resistance (a.k.a., high voltage output or “1”).

While undoped material **116** and doped material **118** can function similarly, their properties can vary from one another. For example, undoped material **116** can have a lower (electrical) resistance drift coefficient (in both the amorphous phase and the polycrystalline phase) and a lower electrical resistance compared to doped material **118**, which has a higher (electrical) resistance drift coefficient (in both the amorphous phase and the polycrystalline phase) and a higher electrical resistance. However, the resistance drift coefficients and electrical resistances of both undoped material **116** and doped material **118** are lower in the polycrystalline set phases than in the amorphous reset phases, respectively. Therefore, the resistance drift coefficient of doped material **118** in the set phase can be lower than the resistance drift coefficient of undoped material **116** in the reset phase. Similarly, the resistance of doped material **118** in the set phase can be lower than the resistance of undoped material **116** in the reset phase.

The result of these properties is shown in FIG. **2**, which is graph **200** of resistance drift coefficient versus resistance between the set and reset phases for various PCM cell configurations. In the illustrated embodiment, resistance drift coefficient is displayed using a linear scale whereas resistance is displayed using a logarithmic scale. The three illustrated PCM cell configurations are homogenous PCM material without a projection liner, homogenous PCM material with a projection liner, and composite PCM section **114** with PL **112** (i.e., PCM cell **100**). For each configuration, there is a line that extends from a set state (without an amorphous zone **126**) to a reset state (includes a large amorphous zone **126**). In the illustrated embodiment, a PCM cell with homogenous PCM material but no projection liner has low resistance and low drift in the set state, and high resistance and high drift in the reset state. A PCM cell with homogenous PCM material and a projection liner has moderate resistance and moderate drift in the set state, and high resistance and low-moderate drift in the reset state. However, PCM cell **100** has low-moderate resistance and low drift in the set state, and high resistance and low-moderate drift in the reset state. Therefore, on average, the resistance drift of PCM cell **100** is less than the other two configurations.

This phenomenon can be explained by the configuration of PCM cell **100**. When all of PCM section **114** is in the polycrystalline set phase, current will primarily travel from heater **108** to top electrode **122** via undoped material **116** due to its lower resistance. Since the resistance drift coefficient of undoped material **116** in the set phase is low, the overall drift of PCM cell **100** will also be low. However, when there is a significant amorphous zone **126** (shown in FIG. **1**), the current can spread out in PL **112** to avoid zone **126** due to its high resistance. Instead, the current will primarily travel to top electrode **122** via doped material **118**. Although the resistance drift coefficient of doped material **118** is higher than that of undoped material **116** (in the same phases), most of doped material **118** is in the polycrystalline set phase (as opposed to the amorphous reset phase of a substantial portion of undoped material **116**). Thereby, the overall drift of PCM **100** will be low-moderate. In addition, the span of resistance between the set state of PCM cell **100** and the reset state of PCM cell **100** is lengthened, which indicates a broader dynamic (i.e., usable) range of resistances. This increases the resolution of PCM cell **100**.

FIG. **3** is a flowchart of method **300** of manufacturing PCM cell **100**. FIGS. **4A-4G** are a series of cross-section views of method **300** of manufacturing the PCM cell. FIGS. **3** and **4A-4G** will now be discussed in conjunction with one another wherein each operation of method **300** is illustrated by one of FIGS. **4A-4G**. In addition, during this discussion, references may be made to features of PCM cell **100** shown in FIGS. **1A-2**, however, some features may be omitted for the sake of simplicity (e.g., bottom wire **102**, bottom electrode **104**, insulator **106**, and top wire **124**).

In the illustrated embodiment, method **300** starts at operation **302**, wherein insulating layer **330** is formed on bottom electrode **104** and insulator **106**. At operation **304**, a via is formed in insulating layer **330**, for example, using etching to form insulator **110**. Then the via is filled to form heater **108**. At operation **306**, PL **112** is formed on heater **108** and insulator **110**, and doped layer **332** is formed on PL **112**. At operation **308**, via **334** is formed in doped layer **332** to expose PL **112** and form doped material **118**. At operation **310**, undoped layer **336** is formed on PL **112** and doped material **118**. At operation **312**, chemical mechanical polishing (CMP) is performed to remove the excess undoped

PCM material to form undoped material **116** that is coterminous with doped material **118** (thus forming PCM section **114**). Then, top electrode **122** is formed on doped material **116** and undoped material **118**. At operation **314**, insulating layer **338** is formed on top electrode **122**.

The components, configuration, and operation of PCM cell **100** and method **300** allow for PCM section **114** to have a composite configuration that is comprised of two different PCM materials (i.e., undoped material **116** and doped material **118**). Thereby, PCM section **114** includes different regions that have different material properties.

FIG. **5** is a flowchart of method **400** of manufacturing PCM cell **100**. FIGS. **6A-6G** are a series of cross-section views of method **400** of manufacturing the PCM cell. FIGS. **5** and **6A-6G** will now be discussed in conjunction with one another wherein each operation of method **400** is illustrated by one of FIGS. **6A-6G**. In addition, during this discussion, references may be made to features of PCM cell **100** shown in FIGS. **1A-2**, however, some features may be omitted for the sake of simplicity (e.g., bottom wire **102**, bottom electrode **104**, insulator **106**, and top wire **124**).

In the illustrated embodiment, method **400** starts at operation **402**, wherein insulating layer **440** is formed on bottom electrode **104** and insulator **106**. At operation **404**, a via is formed in insulating layer **438**, for example, using etching to form insulator **110**. Then the via is filled to form heater **108**. At operation **406**, PL **112** is formed on heater **108** and insulator **110**, and undoped layer **442** is formed on PL **112**. At operation **408**, undoped material **116** is formed from undoped layer **442**, which exposes PL **112**. At operation **410**, doped layer **444** is formed on PL **112** and undoped material **116**. At operation **412**, chemical mechanical polishing (CMP) is performed to remove the excess doped PCM material to form doped material **118** that is coterminous with undoped material **116** (thus forming PCM section **114**). Then, top electrode **122** is formed on doped material **116** and undoped material **118**. At operation **414**, insulating layer **446** is formed on top electrode **122**.

The components, configuration, and operation of PCM cell **100** and method **300** allow for PCM section **114** to have a composite configuration that is comprised of two different PCM materials (i.e., undoped material **116** and doped material **118**). Thereby, PCM section **114** includes different regions that have different material properties.

FIGS. **7A-7H** are cross-section views of alternate embodiment mushroom PCM cells **500A-500H**, respectively. In the illustrated embodiment of PCM cell **500A**, undoped material **516A** tapers towards PL **512A**. Among other methods, PCM cell **500A** can be manufactured using method **300** (shown in FIG. **3**).

In the illustrated embodiment of PCM cell **500B**, undoped material **516B** tapers towards top electrode **522B**. Among other methods, PCM cell **500B** can be manufactured using method **400** (shown in FIG. **5**).

In the illustrated embodiment of PCM cell **500C**, undoped material **516C** tapers towards the middle, creating an hour-glass shape. Among other methods, PCM cell **500C** can be manufactured by forming the bottom and top halves of PCM section **514C** separately. For example, operations **402-410**, and the CMP portion of **412** can be performed to form the bottom half of PCM section **514C**. Then, operations **306-314** can be performed to form the top half of PCM section **514C** and complete PCM cell **500C**.

In the illustrated embodiment of PCM cell **500D**, undoped material **516D** tapers from the middle, creating a diamond shape. Among other methods, PCM cell **500D** can be manufactured by forming the bottom and top halves of PCM

section **514D** separately. For example, operations **302-310**, and the CMP portion of **312** can be performed to form the bottom half of PCM section **514D**. Then, operations **406-414** can be performed to form the top half of PCM section **514D** and complete PCM cell **500D**.

In the illustrated embodiment of PCM cell **500E**, undoped material **516E** tapers towards PL **512E** and only extends halfway through PCM section **514E** therefrom. Among other methods, PCM cell **500E** can be manufactured by forming the bottom and top halves of PCM section **514E** separately.

In the illustrated embodiment of PCM cell **500F**, undoped material **516F** tapers towards top electrode **522F** and only extends halfway through PCM section **514F** therefrom. Among other methods, PCM cell **500F** can be manufactured by forming the bottom and top halves of PCM section **514F** separately.

In the illustrated embodiment of PCM cell **500G**, PCM section **514G** has a graded transition between the centrally-located undoped material **516G** and the peripherally-located, fully doped material **518G**. In other words, the amount of dopant material in PCM section **514G** gradually increases from the center (which has none) to the lateral exterior (which has the full concentration of dopant material). This is in contrast to the abrupt transitions between the undoped material and the doped material present in PCM cells **100**, **500A-500F**, and **500H-500L**. In those embodiments, the amount of dopant material changes in a virtually discontinuous manner since they include composite PCM sections comprised of a portion of undoped PCM material (that is itself homogenous) in contact with a portion of fully-doped PCM material (which is itself homogenous but different from the undoped portion). Among other methods, PCM cell **500G** can be manufactured by performing operations **402-406**. Then, a block mask is placed over the center of undoped layer **442** (which will become undoped material **516G**), and ion implant of dopant is performed (creating doped material **518G**). In some embodiments, undoped material **516G** is actually not doped, and in some embodiments, undoped material **516G** is less doped than doped material **518G**.

In the illustrated embodiment of PCM cell **500H**, PCM cell **500H** is similar to PCM cell **100** (shown in FIG. **1A**). However, PCM cell **500H** lacks a projection liner.

FIGS. **8A** and **8B** are cross-section views of alternate embodiment confined PCM cells **500I** and **500J**, respectively. In the illustrated embodiment of PCM cell **500I**, PCM cell **500I** includes bottom electrode **504I**, PL **512I**, PCM section **514I** (which includes undoped material **516I** and doped material **518I**), and top electrode **522I**. While PCM cell **500I** has a confined cell configuration, it still has doped material **518I** surrounding undoped material **516I** such that doped material **518I** is offset from axis **528I** in a lateral direction (i.e., perpendicular to axis **528I**).

In the illustrated embodiment of PCM cell **500J**, PCM cell **500J** is similar to PCM cell **500I**. However, undoped material **516J** is in direct contact with PL **512J** instead of having doped material **518J** therebetween.

FIG. **9** is a cross-section view of alternate embodiment pillar PCM cell **500K**. In the illustrated embodiment of PCM cell **500K**, PCM cell **500K** includes bottom electrode **504K**, PL **512K**, PCM section **514K** (which includes undoped material **516K** and doped material **518K**), and top electrode **522K**. While PCM cell **500K** has a pillar cell configuration, it still has doped material **518K** surrounding undoped material **516K** such that doped material **518K** is offset from axis **528K** in a lateral direction (i.e., perpendicular to axis **528K**).

FIG. **10A** is a cross-section view of an alternate embodiment bridge PCM cell **500L**, and FIG. **10B** is a top view of

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bridge PCM cell 500L. In the illustrated embodiment of PCM cell 500L, PCM cell 500L includes first electrode 504L, PL 512L, PCM section 514L (which includes undoped material 516L and doped material 518L), and second electrode 522L. PCM cell 500K has a bridge cell configuration such that first electrode 504L is coplanar with second electrode 522L. Thereby, PL 512L and PM section 514L extend parallel to axis 528L (which extends between electrodes 504L and 522L), instead of being centered on axis 528L. During operation of PCM cell 500L, current flows perpendicular to axis 528L proximate to the wide electrodes 504L and 522L. However, as the current crosses the narrow bridge 548L, axis 528L is still parallel to the direction of current flow. Because the active portion of PM section 514L (i.e., the portion where amorphous zone 126 would be created and eliminated) is located in bridge 548L, PCM cell 500L still has doped material 518L surrounding undoped material 516L such that doped material 518L is offset from the flow of current in a lateral direction (i.e., perpendicular to axis 528L).

The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

What is claimed is:

1. A phase change memory (PCM) cell comprising:
  - a first electrode comprised of a first electrically conductive material;
  - a second electrode comprised of a second electrically conductive material; and
  - a phase change section positioned between the first electrode and the second electrode, the phase change section comprising:
    - a first phase change material having a first resistance drift coefficient; and
    - a second phase change material having a second resistance drift coefficient that is greater than the first resistance drift coefficient,
 wherein an axis of the PCM cell extends between the first electrode and the second electrode, and  
 wherein the second phase change material is offset from the first phase change material in a direction that is perpendicular to the axis.
2. The PCM cell of claim 1, wherein the phase change section has a graded transition between the first phase change material and the second phase change material.
3. The PCM cell of claim 1, wherein the phase change section has an abrupt transition between the first phase change material and the second phase change material.
4. The PCM cell of claim 1, wherein the first phase change material is undoped and the second phase change material is doped.
5. The PCM cell of claim 1, further comprising: a projection liner between the first electrode and the phase change section, wherein the projection liner is comprised of a material with a resistivity between a resistivity of a polycrystalline phase of the first phase change material and a resistivity of an amorphous phase of the first phase change material.

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6. The PCM cell of claim 1, further comprising: a heater positioned between the first electrode and the phase change section.

7. The PCM cell of claim 6, wherein,

the heater has a first width,

the phase change section has a second width, and

the second width is greater than or equal to thrice the first width.

8. The PCM cell of claim 6, further comprising: a projection liner between the heater and the phase change section, wherein the projection liner is comprised of a material with a resistivity between a resistivity of a polycrystalline phase of the first phase change material and a resistivity of an amorphous phase of the first phase change material.

9. A method of manufacturing a phase change memory (PCM) cell, the method comprising:

forming a first electrode;

forming a phase change section electrically connected to the first electrode, wherein the phase change section comprises:

a first phase change material having a first resistance drift coefficient; and

a second phase change material having a second resistance drift coefficient that is greater than the first resistance drift coefficient; and

forming a second electrode on the phase change section, wherein an axis of the PCM cell extends between the first electrode and the second electrode, and

wherein the second phase change material is offset from the first phase change material in a direction that is perpendicular to the axis.

10. The method of claim 9, further comprising: forming a projection liner between the first electrode and the phase change section.

11. The method of claim 9, further comprising: forming a heater between the first electrode and the phase change section.

12. The method of claim 11, wherein

the heater has a first width,

the phase change section has a second width, and

the second width is greater than or equal to thrice the first width.

13. The method of claim 9, wherein forming the phase change section comprises:

depositing the second phase change material;

etching the second phase change material to form a via; and

filling the via with the first phase change material.

14. The method of claim 9, wherein forming the phase change section comprises:

depositing the first phase change material;

etching the first phase change material to form a pillar; and

surrounding the pillar with the second phase change material.

15. The method of claim 9, wherein forming the phase change section comprises: grading the phase change section to transition from the first phase change material to the second phase change material.

16. The method of claim 9, wherein the first phase change material is undoped and the second phase change material is doped.

17. A phase change memory (PCM) cell comprising:

a first electrode comprised of a first electrically conductive material;



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- a second electrode comprised of a second electrically conductive material;
- a phase change section positioned between the first electrode and the second electrode, the phase change section comprising:
- 5 a first phase change material having a first resistivity; and
- a second phase change material having a second resistivity that is greater than the first resistivity; and
- 10 a projection liner between the first electrode and the phase change section, wherein the projection liner is comprised of a material with a resistivity between a resistivity of a polycrystalline phase of the first phase change material and a resistivity of an amorphous phase of the first phase change material,
- 15 wherein an axis of the PCM cell extends between the first electrode and the second electrode, and
- wherein the second phase change material is offset from the first phase change material in a direction that is perpendicular to the axis.
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- 18.** The PCM cell of claim 17, wherein:
- the first phase change material has a first resistance drift coefficient; and
- a second phase change material having a second resistance drift coefficient that is greater than the first
- 25 resistance drift coefficient.
- 19.** The PCM cell of claim 17, further comprising a heater positioned between the first electrode and the phase change section.

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